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SUPERSEDING
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MILITARY SPECIFICATION

PALLET LOAD ASSEMBLY FOR SONOBUOYS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for unit loads of sonobuoys packaged in containers.

1.2 Title Definition. The "load" shall mean sonobuoys, acoustic communication sonobuoys, bathythermograph transmitters, shipped in sonobuoy launcher-containers and/or shipped in sonobuoy shipping and storage container CNU-239/E or equivalent. The "pallet load assembly" shall mean the load, the pallet, and the cap or other protective appendages secured together in a unitized assembly. (See 6.5 for other definitions)

1.3 Classification. The pallet assembly covered by this specification shall be of the following: (6.2(a))

Type I. For sonobuoy shipping and storage containers CNU-239/E interlocked and grouped, base 1.118 X 1.118 m (44" X 44") height 1.321 m (52") maximums.

Type II For sonobuoy launcher-containers individually interlocked with the pallet base, base .864 X .864 m (34" X 34") height 1.245 m (49") maximums.

Load Plan A. Full load of 36, all of one RF channel, or all muti-RF channel.

B. Full load of 36, equal distribution of 12 sets of three RF channels.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733 by using the self-addressed Standarization Documentation Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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C. Full load of 36, equal distribution of three sets of 12 RF channels and six sets of acoustic channels.

D. Short load of 31, one set of 31 RF channels

E. Short load of 32, one set of 31 RF channels plus one (sequentially different in each successive load for an equal distribution of eight sets of four acoustic channels).

F. Full mixed load of 36. 1/

G. Partial load (see.3.4.7.3). 1/

H. Small load (see 3.4.7.4). 1/

Class (see 3.4.8.2)

1. Non Flammable.

2. Delayed Ignition

3. Fire Resistant, self extinguishing.

4. Flammable, but no more flammable than load.

5. Flammable, wood limit.

Level

B. Limited shipment and reshipment, protected long term storage, and unrestricted deployed short term storage.

C. Commercial safe delivery direct to consumer. (See Load Plan H.)

1.4 Identification. The pallet load assembly shall be identified by manufacturer's Data List number that identifies the complete configuration and revision thereto, to the piece part level. (3.4.9.4)

1.5 Associated Equipment. Standard commercial and military cargo and warehouse pallet handling equipment including fork lifts, wheeled hand lifts, and slings.

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents. - The following documents, of the issue in effect on the date of inviation for bids or request for proposal, form a part of this specification to the extent specified herein.

NOTE:

1/ Specific purpose shall be included in order. (6.2(c))

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SPECIFICATION

FEDERAL

NN-P-71 - Pallet, Material Handling, Wood, Stringer Construction, 2-Way And 4-Way (Partial)

QQ-S-781 - Strapping, Steel, and Seals

MILITARY

MIL-P-3938 - Pallet, Material Handling, Hardwood, Stringer Construction, 4-Way (Partial)

MIL-L-81745(AS) - Launcher-Container; "A" Size Stores (SLC), General Specification For

STANDARDS

FEDERAL

Fed. Test Method Std 101 - Preservation, Packaging, and Packing Materials: Test Procedures

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-810 - Environmental Test Methods

PUBLICATIONS

NAVAIR 28-SSQ-500-1 - Sonobuoy Instructional Manual

DRAWINGS

80206 TE21077 - "A" Size Store Launch System

2.2 Availability of Documents

(a) When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable specifications and standards required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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(b) The drawing listed is available on the same basis, from Naval Air Development Center, Warminster, Pennsylvania 18974, Attn: 60313.

2.3 Precedence. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specifications.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications.

(c) Referenced Specification, Standards, and Requirements - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

(d) Drawings - Drawings or lists of drawings referenced herein shall have precedence over all subsidiary drawings. All referenced drawings shall apply to the extent specified thereon.

3. REQUIREMENTS

3.1 Classification of Requirements. The requirements for the pallet load assembly are classified herein as follows:

REQUIREMENTS	PARAGRAPH
First Article	3.2
Design Changes	3.3
Design and Construction	3.4
Operational Environment	3.5
Performance	3.6
Safety	3.7
Reliability	3.8

The referenced tests follow each requirement. All tests are found in tables at the end of this specification. They are referred to by table number, increment or group, and sequence number. Example: (see II-A-1).

3.2 First Article. When specified in the order, the Contractor shall furnish sample units for First Article inspection and approval; see 6.4. First Article samples shall be fabricated using the same tooling, methods, and procedures that will be used in production. The sample shall successfully complete all tests under the First Article Quality Assurance provisions herein prior to a release to production; see 4.5. Each configuration and maximum load weight shall be qualified by test. Loads of less weight but of the same configuration are qualified without additional testing. Each different configuration to be used shall be qualified by test. Configuration differences shall include the following:

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Change in material and increase in the use of reclaimed materials.
Change in design
Increase in load weight
Decrease in materials dimensions
Change in load orientation
New or increased size of material defects that are acceptable.
Addition of holes constructed or open corners in the load.

3.3 Design Changes. All changes in design that may affect the performance, reliability or interface shall be tested by the procuring activity prior to approval for use in production. The pallet load assembly, manufactured with the changes incorporated, shall undergo First Article Testing, all or in part appropriate to the characteristics affected by the change as specified under the Quality Assurance provisions herein; see 4.8.

3.4 Design and Construction. The design of the pallet load assembly (less the load) shall be detailed in specification control drawings provided under the contract (6.3) or prior contracts. The details shall include: assemblies; subassemblies; piece parts; material; preservation; fire retardation treatment; dimensions; securing devices (including nails and staples), their number and location; applicable sampling test requirements for strength holding power and construction of the pallet load assembly which shall enable it to reliably (3.8) perform as specified (3.6) under all operational environments expected (3.5) and not become a safety hazard to material and personnel (3.7). (see test II-D-1 and V-1, 2, and 5).

3.4.1 Construction. The material and processes used shall be the lowest in cost that will meet the requirements of this specification and will allow high production rates to be used. Reclaimed material shall be used where practical without affecting reliability and safety. The construction shall meet the requirements of one-way, one-time use for Type I and a minimum of one reuse cycle for Type II. The assembly of the unitized load shall be complete and shall contain the following elements: 2/

(a) Pallet, with 4-way (partial) entry, double wing; or equivalent base unit. (6.5) (see test II-D-2(a)).

(b) Cap, or equivalent top unit. (see test II-D-2(b))

NOTE:

2/ The ideal sonobuoy stowage, handling, and shipping position is with the long axis vertical,decelerator up. In this position the major transportation/handling shock is sustained within the sonobuoy design limits of 100 g's for 11 ms. The design provides for major shock of launching, water entry, and transportation/handling when applied in coincidence. This is also the optimum position for o-ring seals to prevent out-of-round set and for plastic store launcher-containers to prevent out-of-round set with age. It should be noted that shocks in excess of 50 g's for 11 ms applied from other than the designed direction may cause sonobuoy malfunction.

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(c) Load security (see tests II-D-2(c), 4.a, 8.b, and 18)

Type I.

Collar (or fence), non-flexible, top and bottom; or equivalent side units. The collars shall prevent unlocking of the interlocked containers and groups of containers; and shall prevent the load from shifting from the pallet or cap.

Type II

Base interlock receptacles that secure each sonobuoy launcher-container breech locking lugs and prevent unlocking by the weight of each unit detenting its rotational position. Alternately, a base load unit item organizer, pockets, or sockets that prevents each sonobuoy launcher-container individually from tip-over. Cap organizer that maintains each sonobuoy launcher-container in its "base" relative position (such as a foam blanket, a formed piece, or other restraint). Rigid side panels that support the cap and overstacking without allowing the overstacking weight to bear on the load, but are individually removable or collapsible. A cap and base collar, clips, or receptacles may be used to restrain the side panels; mailing or other permanent/semi-permanent means shall not be used. Cap clearance and superimposed load support may be insured by the addition between the base and the cap of one or more free struts interposed between launcher-containers and/or in the corners of the load. The alternative to cap clearance of load is performance (proven by test) that will prevent superimposed load shearing damage to the launcher-containers due to inertia during van and rail shipment.

(d) Strapping, or equivalent securing devices, to secure the load assembly together. Strapping shall conform to QQ-S-781, Class 1, Type I Heavy Duty; and seals shall conform to QQ-S-781, Type D, Style optional, Class H. Equivalent securing devices shall provide the same load securing reliability and resistance to corrosion. (see test II-D-2(d)).

(e) Hardware as required; such as: corner, chafing plates, nailing plates, etc.

(f) (Type I only) Marking placards, or equivalent. (see test II-D-2(e)).

(g) (Type I only) Fillers to make up full load, if needed. (see test II-D-2(g)).

(h) (Type I only) Cap supports, if needed. (see test II-D-2(f)).

(i) Drainage provision, such as will prevent water standing in any part of the assembly. (see test II-D-2(g)).

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3.4.2 Form Factor. The following load, stowage, and handling equipment interface provisions shall be included in the design:

(a) Load layout: (see test II-D-2(g)).

Type I.

36 sonobuoy shipping and storage containers CNU-239/E (or equivalent), interlocked, grouped, and strapped. The load shall not extend over (overhang) the side of pallet and cap units, nor shall it underhang except for collar (or fence) and assembly clearance.

Type II.

36 sonobuoy launcher-containers LAU-112A and similar types that do not require an exterior shipping container. The load shall extend with minimum clearance to the exterior panels.

(b) Exterior assembly; length, width, and height see 1.3 (see test II-D-2(h).

(c) Load interface: (see test II-D-2(n) and IV-1a).

Type I.

Load bearing surface; each individual sonobuoy shipping and storage container CNU-239/E (or equivalent) shall have 100% of the surface area bearing against its two ends without holes, steps, or ridges. This shall include the area of contact developed by the overstacking requirement and the maximum deformation resulting on the ends.

Type II.

The base to sonobuoy launcher-container breech end interface shall conform to Figure 6 of MIL-L-81745(AS). The base bearing shall support the weight of the launcher-container, with its assembled sonobuoy, without causing deformation of the breech or locking lugs. The cap to muzzle end interface shall similarily conform without causing breakout cap failure or deformation due to overstacking.

(d) Fork lift and wheeled hand lift entries; the provisions for entry shall be equivalent to those illustrated in Figures 1 and 2 of MIL-P-3938 (Reference).(see test II-D-2(a)).

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(e) Pallet bar sling entries; the wing provisions for hoisting slings shall be equivalent to that illustrated in Fig. 1, Side View of MIL-P-3938 (Reference).(see test II-D-(a)).

(f) Top and bottom bearing surface; the pallet load assembly shall have bearing surfaces to rest on and to support other pallet load assemblies. They shall be within parallel limits that will allow overstacking (3.4.4). (see test II-D-2(j)).

3.4.3 Weight The pallet load assembly shall not exceed a total weight of 807.2 kgm (2000 lbs). (see test II-D-2(k)).

3.4.4 Center of Gravity. The center of gravity shall be within the limits that will allow stacking four pallet load assemblies, one centered on top of the other, on a 5° inclining surface without loss of stability. (see test II-D-11).

3.4.5 Breakdown Load Inclination (Type II only). The separate individual units of the load (sonobuoy launcher-containers) shall be restrained in place when the load base is out of the horizontal up to (and including) 30° in any direction, with all group and load securing straps (or other devices) removed and the cap removed. (see test II-D-13).

3.4.6 (relocated see 3.4.7.6)

3.4.7 Load construction:

3.4.7.1 Type I Load. The load shall be banded together without damage to the sonobuoy shipping and storage container CNU-239/E or other similar sonobuoy container. Only sonobuoy containers of the same height and matching interlock features (or lack thereof) shall be combined in one load. The top and bottom of the containers may be compressed slightly if designed with spring ends as is the CNU-239/E. The sides shall not be deformed. (see tests II-D-2(1) and 12).

(a) Sonobuoys that weigh over 11.3 kgm (25 lbs) shall have their containers interlocked and banded together in subgroups of two. See Figure 1. (see test V-3)

(b) All containers, less than 11.3 kgm (25 lbs) shall be interlocked and banded together in groups of four. Subgroups of two shall also be combined in groups of four in a manner that will insure stability when standing on end, when not unitized (3.6.6). See Figure 1. (see test V-3).

(c) Groups and subgroups of containers shall be placed on the pallet or base with the cap of each container up. Normally this will place the sonobuoy base plate down and launch-

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container breech down. Except this may be reversed to comply with the center of gravity requirement (3.4.4) which shall determine which end of the container is up. 3/ All groups shall be interlocked and banded together. There shall be three groups to a side when consolidated in this manner, and one in the middle. See Figure 1. (see test V-3).

(d) The load shall support a cap or other unit that will in turn support and distribute the weight of overstacked pallet loads.

3.4.7.2 Type II Load. The load shall be banded together or otherwise locked in place without preloading the sonobuoy launcher-container, except for pallet cap cushion (blanket) or organizer compression. The cap shall be secured down tight onto the side panels to form a rigid box. The individual units of the load, the sonobuoy launcher-containers, shall be individually locked to the pallet base or supported by base organizers, sockets, or nesting; and cushioned, organized, or socketed against movement at the cap. When locked, they shall be breech down. They shall not be banded in groups or directly as a load. (see II-D-2 (m) and 18; V-4).

3.4.7.3 Partial Loads. (see test II-D-2(n))

Type I

Any method may be used to construct a partial load by the use of empty containers, spacers, fillers, or constucted holes provided that the method was qualified under the same conditions the full load was qualified. Full superimposed load (3.5.3) requirements, shall apply. The most cost effective method should be used. The constructed hole method is suggested. Three containers must be interlocked in an "L" pattern. The open part of the "L", or the hole, shall not be placed adjacent to another hole to insure that the load interlock will be maintained. The first holes to be utilized should be the corners. A block of wood (or other) may be used to maintain separation of the open part. See Figure 1.

Type II

Distribute the number required to provide an evenly balanced load. Full superimposed load (3.5.3 and 3.6.6) requirements shall apply.

NOTE:

3/ It should be noted that the sonobuoy may fail due to its design requirements of only 50 g's in the reversed direction (100 g's if not reversed).

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3.4.7.4 Small Loads. Small quantities of containers, for which no qualified load was developed under this specification, and limited to the delivery of the final lot of a contract or units for test shall be boxed and if the quantity warrants, palletized in the horizontal position; in which case, the cap shall be supported by structure other that the load. They shall be protected for delivery in Continental U.S., Level C. The methods selected shall be approved by the Government representative at the point of origin.

3.4.7.5 RF and Acoustic Channel Load Plan. The distribution of radio frequency channels in a load shall be in accordance with the load plan specified in the order (6.2(b)). See 1.3 and Table I.

3.4.7.6 Lot and Contract Distribution. Unless directed otherwise by the procuring activity more that one lot and more that one contract of one type of sonobuoy may be mixed in one unitized load for shipping purposes, providing the load plan (3.4.7.5) is complied with and the marking (3.4.9.1(c) and (e)) indicates the contents.

3.4.8 Material Treatment:

3.4.8.1 Preservation. The preservation required is limited to that which protects against the environment (3.5) and assures performance (3.6). The surface of metal parts, not including banding and seals, may rust or oxidize providing the strength or holding power is not reduced. Wood may discolor and develop minor cracks providing it does not develop major cracks or rot. Plastic may discolor providing that it does not become embrittled or develop cracks. All markings shall remain legible and marking surfaces shall remain intact. See 3.4.1(d) for strapping and seal requirements. (see test II-C-1 and 5, and V-1). (See 6.3 Certification (i)).

3.4.8.2 Fire Protection. The order (6.2(b)) specifies the class. Additives shall be added or surfaces treated or finished, as required, to obtain the level of fire protection for the class specified, but only to add more protection. (See 1.3)(see test II-C-6) (See 6.3 Certification (i)).

Class 1 Non-Flammable or Fire Resistant, will not burn in a fuel fire.

Class 2 Delayed ignition, non-flammable for first five minutes, flammable thereafter.

Class 3 Self Extinguishing within five minutes on removal of applied flame.

Class 4 Fire Retardent, slow burning, but no more flammable than the load (the sonobouy launcher-containers and/or the shipping and storage container CNU-239/E) and self extinquishing rate is equal to or better than the load.

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